

Key-Value Stores and Graph Databases

Key-Value Stores

Description

Store data as key-value pairs where the key is a unique identifier and the value is the associated data. Highly performant for simple read and write operations.

Example: Redis

Use Cases: Caching, session management, real-time analytics

Data Structure

Key-value stores support multiple value types:

- **String:** Simple key to string mapping
- **Hash:** Key to field-value map (for structured objects)
- **List:** Key to ordered list of values

Code Example

```
SET "user:12345:name" "John Doe"
SET "user:12345:age" "30"

HSET "user:12345:address" "street" "123 Main St"
HSET "user:12345:address" "city" "Anytown"
HSET "user:12345:address" "state" "CA"
HSET "user:12345:address" "zip" "12345"

LPUSH "user:12345:hobbies" "reading"
LPUSH "user:12345:hobbies" "hiking"
LPUSH "user:12345:hobbies" "coding"
```

Graph Databases

Description

Store data in graph structures. Designed to handle complex and interconnected data with high efficiency.

Example: Neo4j

Use Cases: Social networks, recommendation systems, fraud detection

Core Concepts

- **Nodes** - Represent entities (e.g., users, posts)
- **Relationships** - Represent connections between entities (e.g., friendships, likes)
- **Properties** - Attributes of nodes and relationships (e.g., name, age, post content)

Example: Social Network

Entities:

- Users: Alice, Bob, Carol
- Posts: Created by users
- Relationships: Friendships between users and likes on posts

Key Advantage Over Relational DBs

Graph databases traverse relationships natively without expensive JOIN operations, making them significantly faster for highly connected data queries.

Quick Comparison

Feature	Key-Value Store	Graph Database
Data Model	Key-Value pairs	Nodes and Edges

Feature	Key-Value Store	Graph Database
Best For	Simple lookups, caching	Complex relationships
Example	Redis	Neo4j
Query Speed	Very fast for direct access	Fast for relationship traversal
Use Case	Sessions, caching	Social networks, fraud detection